Open Book Quiz Session 6: Applied Biostatistics Part 2 ...

Attempt 2 of 3

Written Oct 21, 2026 2:47 PM - Oct 21, 2026 2:55 PM

Attempt Score **15 / 15 - 100 %**

Overall Grade (Highest Attempt) **15 / 15 - 100 %**

Question 1 1 / 1 point

A case-control study examines alcohol (≥ 2 drinks/day) and lung cancer. The crude odds ratio is 1.8 (95% CI 1.4 to 2.3). After adjusting for pack-years of smoking in a logistic regression, the adjusted odds ratio is 1.05 (95% CI 0.85 to 1.30). What is the best interpretation?

- ○ Alcohol is an independent risk factor for lung cancer
- ○ The adjusted result must be a Type II error because adjustment reduces power
- ✓ ◉ Smoking confounded the crude association; after accounting for it, no significant association remains
- ○ Smoking is an effect modifier of the alcohol–lung cancer association
- ○ Adjusting for smoking introduced bias into the estimate

▼ Hide question 1 feedback

Feedback

The estimate moved from 1.8 to about 1.0 once smoking was included, and the adjusted CI includes 1. That pattern is confounding. Effect modification would

mean the association differs across smoking strata, which a single adjusted OR does not show.

Question 2

1 / 1 point

A cohort study of 20,000 adults finds a Pearson correlation of r = 0.01 (p = 0.16) between BMI and all-cause mortality. A plot of mortality by BMI category shows higher mortality at both very low and very high BMI, with the lowest mortality at a BMI of about 22–25. What is the best conclusion?

- ○ Higher BMI protects against death
- ✓ ◉ Pearson r measures only linear association; a J-shaped relationship can produce an r near 0
- ○ A Spearman correlation would necessarily show a strong positive relationship
- ○ The study was too small to detect a relationship
- ○ BMI is unrelated to mortality

▼ Hide question 2 feedback

Feedback

Risk rises at both ends, so a straight line fits poorly and r is near 0 even though the relationship is strong. Always look at the plot before interpreting r.

Question 3

1 / 1 point

A cohort study follows 8,000 postmenopausal women for up to 10 years until their first hip fracture; women lost to follow-up are censored. Long-term proton pump inhibitor use has an adjusted hazard ratio of 1.30 (95% CI 1.10

to 1.54), adjusted for age, BMI, and corticosteroid use. Which statement is correct?

- ○ Logistic regression was used; PPI users had 30% higher odds of fracture
- ○ Linear regression was used; PPI users had their fracture 1.3 years earlier
- ✓ ◉ Cox proportional hazards regression was used; PPI users had a 30% higher hazard of hip fracture at any point during follow-up, and the association is statistically significant
- ○ Cox regression was used; the association is not significant because the CI does not include 0
- ○ Kaplan–Meier analysis was used; 30% of PPI users had a fracture

▼ Hide question 3 feedback

Feedback

Time-to-event data with censoring and covariates call for Cox regression, which reports hazard ratios. The null value for an HR is 1, and the CI excludes 1, so the association is significant. This links to the survival analysis in Sessions 3–4.

Question 4 1 / 1 point

A linear regression developed in adults aged 40 to 70 years predicts eGFR from age: eGFR = 130 − 0.9 × age. A clinician uses it to estimate eGFR for a 22-year-old and a 95-year-old. What is the main problem?

- ✓ ◉ Extrapolation beyond the range of data used to build the model
- ○ Confounding
- ○ The slope is negative
- ○ R^2 was not reported

- ○ Type I error

▼ Hide question 4 feedback

Feedback

A regression equation describes the relationship only within the range of the data. Outside ages 40 to 70 the relationship may be different, so these predictions are unreliable

Question 5 1 / 1 point

A healthy 25-year-old has a 12-test chemistry panel. Each test's reference range is the central 95% of healthy people. Assuming the tests are independent, what is the probability that at least one result is flagged as abnormal?

- ○ 5%
- ○ 34%
- ○ 26%
- ✓ ◉ 46%
- ○ 12%

▼ Hide question 5 feedback

Feedback

P(all 12 normal) = $0.95^{12} \approx 0.54$, so P(at least one flagged) ≈ 1 − 0.54 = 0.46. A single mildly abnormal value in a healthy person is often expected; interpret it in context.

Question 6

1 / 1 point

Investigators examine the relationship between pain severity on a 0–10 numeric rating scale and daily opioid dose (morphine milligram equivalents) in 120 postoperative patients. Opioid dose is heavily right-skewed with several extreme values. Which measure of association is most appropriate?

- ✔ (●) Spearman rank correlation
- ( ) One-way ANOVA
- ( ) Paired t-test
- ( ) Chi-square test
- ( ) Pearson correlation

▼ Hide question 6 feedback

Feedback

Spearman's ρ uses ranks, so it suits ordinal data, skewed distributions, and outliers. Pearson r assumes two continuous, roughly normal variables and is sensitive to extreme values.

Question 7

1 / 1 point

In 300 adults, waist circumference and CT-measured visceral fat area are correlated: r = 0.70 (p < 0.001). Which statement is correct?

- ( ) The relationship is weak because r is below 0.80
- ( ) A larger waist circumference causes more visceral fat
- ✔ (●) About 49% of the variation in visceral fat area is explained by its linear relationship with waist circumference
- ( ) 70% of the variation in visceral fat is explained by waist circumference

- Each 1-cm increase in waist circumference raises visceral fat area by 0.70 cm^2

Hide question 7 feedback

Feedback

$r^2 = 0.70^2 = 0.49$. r is not a slope (C), correlation does not show causation (D), and r = 0.70 is a strong correlation (E).

Question 8 1 / 1 point

In 600 adults with type 2 diabetes, a linear regression of HbA1c (%) on BMI gives a slope of $\beta = 0.08$ per 1 kg/m^2 (95% CI 0.05 to 0.11) and an intercept of 4.9. Two patients differ in BMI by 10 kg/m^2. What is the predicted difference in their mean HbA1c?

- 8.0 percentage points
- 4.9 percentage points
- 0.11 percentage points
- ✓ 0.8 percentage points
- 0.08 percentage points

Hide question 8 feedback

Feedback

The slope is the change in mean outcome per 1-unit change in the predictor: 10 × 0.08 = 0.8 percentage points. The intercept does not enter a difference between two patients

Question 9

1 / 1 point

A hospital wants to identify which factors predict 30-day readmission (yes/no), examining age, number of medications, primary diagnosis, and discharge day of the week at the same time. Which analysis is most appropriate?

- ○ One-way ANOVA
- ✔ ◉ Multiple logistic regression
- ○ Pearson correlation
- ○ Multiple linear regression
- ○ Cox proportional hazards regression

▼ Hide question 9 feedback

Feedback

The outcome is binary and there are several predictors, so multiple logistic regression is appropriate; it reports adjusted odds ratios. Linear regression is for continuous outcomes and Cox regression for time-to-event outcomes.

Question 10

1 / 1 point

Hemoglobin in healthy adult women is approximately normal, with a mean of 13.5 g/dL and an SD of 1.0 g/dL. A 34-year-old woman has a hemoglobin of 11.0 g/dL. What is her z-score?

- ○ -11
- ○ -5.0
- ○ +2.5
- ○ -1.0

✔ -2.5

Hide question 10 feedback

Feedback

z = (value − mean) ÷ SD = (11.0 − 13.5) ÷ 1.0 = −2.5. She is 2.5 SDs below the mean; under 1% of healthy women are this low.

Question 11

1 / 1 point

In a registry of 50,000 adults, a linear regression of SBP on daily step count gives β = −0.4 mmHg per 1,000 steps (95% CI −0.5 to −0.3), $p < 0.001$, R^2 = 0.02. Which conclusion is best?

- Step count is a strong predictor of SBP because $p < 0.001$
- Each additional 1,000 steps causes a 0.4 mmHg drop in SBP
- 2% of participants had lower SBP when they walked more
- The association is not statistically significant because R^2 is small
- ✔ The association is statistically significant, but step count explains only about 2% of the variation in SBP, and the effect per 1,000 steps is small

Hide question 11 feedback

Feedback

With n = 50,000, even a small, weak association is statistically significant. R^2 = 0.02 means step count explains little of the variation in SBP. Observational data do not establish causation.

Question 12

1 / 1 point

Fasting plasma glucose in healthy adults is approximately normally distributed with a mean of 90 mg/dL and an SD of 8 mg/dL. Approximately what percentage of healthy adults have a fasting glucose above 106 mg/dL?

- ○ 47.5%
- ○ 16%
- ○ 5%
- ○ 32%
- ✓ ◉ 2.5%

▼ Hide question 12 feedback

Feedback

106 mg/dL is the mean + 2 SD. About 95% of values lie within ±2 SD, so 5% lie outside, split equally between the two tails: 2.5% above.

Question 13

1 / 1 point

A cross-sectional study of 5,000 adults finds that people who carry a cigarette lighter have a much higher prevalence of lung cancer than people who do not. What is the most likely explanation for this association?

- ○ Measurement Bias
- ○ Selection Bias
- ✓ ◉ Confounding by Cigarette Smoking
- ○ Type I Error
- ○ Reverse Causation

Hide question 13 feedback

Feedback

Smoking is associated with both carrying a lighter and lung cancer, and it is not on the causal path between them. A confounder can create a strong correlation with no causal link.

Question 14

1 / 1 point

A random sample of 100 patients with hypertension has a mean SBP of 140 mmHg and an SD of 20 mmHg. Which range is the approximate 95% confidence interval for the population mean SBP?

- 138 to 142 mmHg
- 120 to 160 mmHg
- 138 to 142 mmHg
- ✓ 136 to 144 mmHg
- 100 to 180 mmHg

Hide question 14 feedback

Feedback

SE = SD ÷ √n = 20 ÷ 10 = 2, so the 95% CI ≈ 140 ± 1.96 × 2 ≈ 136 to 144 mmHg. The range 100 to 180 (± 2 SD) describes where about 95% of individual patients fall, not the precision of the mean.

Question 15

1 / 1 point

A lipid clinic reports fasting triglycerides for 1,000 adults: mean 185 mg/dL, median 140 mg/dL, mode 120 mg/dL. Which statement best describes the distribution and how to summarize it?

- ✔ (●) Positively skewed; report the median and interquartile range (IQR)
- ( ) Negatively skewed; report the mean and SD
- ( ) Normally distributed; report the mean and SD
- ( ) Positively skewed; report the mode and range
- ( ) Negatively skewed; report the median and IQR

▼ Hide question 15 feedback

Feedback

Mode < median < mean means a long right tail: a few very high values pull the mean up. For skewed data, the median and IQR describe a typical patient better than the mean and SD.

Done